

INSIGHT 2.0 — GRAND FINALE · IASDS STUDENTS' CLUB, UNIVERSITY OF DHAKA

Survival Classification of Lung-Cancer Patients: A Cross-Validation-First Approach on SEER Registry Data

Team NeurAps · 5 September 2026

0.880410

private weighted F1

4th / 79

final private rank

Outline — covering all eight required components

1. **Summary of Our Work** — overall approach and headline result
2. **Exploratory Data Analysis** — imbalance, the diagnosis-year effect, data quality
3. **Feature Selection** — what we kept, what we audited, and why
4. **Feature Engineering** — features created and transformed, with rationale
5. **Model Selection** — models considered and the reasoning behind choices
6. **Model Evaluation** — validation design and metric-faithful evaluation
7. **Model Comparison** — head-to-head performance of every model tried
8. **Best Findings** — key insights and final conclusions

Section 1

Summary of Our Work

1

Problem statement and result

Given SEER cancer-registry records of lung-cancer patients (all primary sites C34.x), predict `vital_status` $\in \{\text{Alive, Dead}\}$. Metric: **weighted F1** on a hidden test set (40% public / 60% private).

24, 000

training records

36, 000

test records (1.5×)

36

clinical features

6/day

submission budget

Our approach in one sentence

Engineered clinical features → a blend of three gradient-boosted tree models, trained twice (plain, then with pseudo-labelled test rows), mixed with 5% each of RandomForest, ExtraTrees and an MLP, thresholded at 0.575 — every decision gated by honest out-of-fold cross-validation.

Section 2

Exploratory Data Analysis

2

The target is heavily imbalanced

83.0%

of patients are Dead

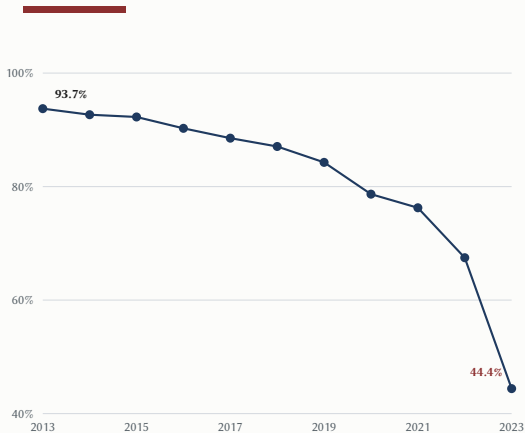
17.0%

are Alive

Why this matters for weighted F1:

- An all-Dead baseline reaches 0.907 Dead-class F1 but only ~ 0.75 weighted F1 — the minority class carries decisive weight.
- The decision threshold becomes a first-class model parameter.
- Stratified resampling is mandatory in cross-validation.

The dominant covariate: year of diagnosis



93.7% → 44.4% — a right-censoring artifact: recent patients have not been followed long enough to die.

- Train/test share the year distribution ⇒ random split ⇒ StratifiedKFold is the honest CV scheme.
- Errors concentrate in recent years (23% in 2023 vs 6% in 2013).
- `years_since_dx` becomes a first-class feature.

Data quality: sentinel codes, not NaN

Coded missingness

Tumor size uses 999 / “Blank(s)” / “Unknown”; one surgery-scope column is 66.5% null by era definition. Decoded to numerics + explicit missingness flags.

Era-dependent columns

Size and staging switch coding systems across eras; we coalesce into single best-available features instead of discarding either era.

Leakage checks

No duplicate IDs; row order uninformative; every high-power feature audited individually.

Informative categories

“Death certificate only” and “died prior to recommended surgery” directly signal the outcome, appear in both splits — legitimate, powerful features.

Section 3

Features: Selection & Engineering

3

Feature selection: what we kept, and why

Starting point: all 36 provided features. Final model: all 36 + ~ 15 engineered. Retention was a **tested** decision, not a default.

- **No hard filtering:** GBDTs select implicitly — uninformative features are never chosen for splits. Dropping low-importance features worsened CV every time we tried.
- **Importance-ranked signal:** surgery-decision codes, summary stage / TNM, year of diagnosis, age, tumor size, metastasis flags — clinically coherent.
- **Leakage audit as selection:** every strong feature individually audited; all passed.
- **No shift-based drops:** train/test distributions align; nothing sacrificed for stability.

Feature engineering

Feature	Construction and justification
t/n/m_ord	T/N/M staging mapped to ordered scales — staging is clinically ordinal
tnm_sum	composite stage score, metastasis double-weighted
tumor_size_best	three era-coded columns parsed and coalesced + missing flags
mets_count	count of metastatic sites — graded severity signal
nodes_ratio	positive/examined nodes — normalizes surgical thoroughness
treatment flags	surgery, radiation, death-certificate-only — physician prognosis signal
years_since_dx	the censoring proxy (see EDA)

Deliberately avoided: fold-external target encoding, anything fitted on test, AutoML.

Section 4

Model Selection & Evaluation

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Model selection: why three GBDTs

- **LightGBM** — leaf-wise growth, histogram splits, native categoricals: the workhorse.
- **XGBoost** — depth-wise growth, second-order gradients: same family, *different mistakes*.
- **CatBoost** — ordered target statistics: built for 114-level histology codes.
- Their OOF correlation is ~ 0.97 — but the 3% disagreement sits exactly on the uncertain rows where F1 is decided; the blend beats every member.
- **Diversity mixture**: RandomForest, ExtraTrees, MLP at 5% each — weaker solo, decorrelated errors; verified $+0.0006$ by split-half testing.
- **Why not deep learning?** Measured, not assumed: MLP 0.874 solo, TabPFN correlated 0.98 with the blend. At 24k rows, trees win.

The final architecture



Model evaluation protocol (identical for every candidate)

Stratified 10-fold out-of-fold weighted F1 with tuned threshold; multiple seeds; honest split-half verification for every *selected* quantity; the public leaderboard used only for controlled single-variable probes.

Pseudo-labelling: our biggest win, done honestly

- The test set is $1.5\times$ the training data: 36,000 unlabelled in-distribution rows.
- Only extreme confidences cross over ($p > 0.99 \rightarrow \text{Dead}$, $p < 0.02 \rightarrow \text{Alive}$): 7,455 rows, expected label noise 1–2%.
- **CV integrity:** pseudo-rows join only the training side of each fold; evaluation stays on original labelled rows — the estimate remains unbiased.
- Extensions tested and rejected: wider bands (no gain), a second round (no gain).

Section 5

Model Comparison & Experiments

5

Model comparison: every model, head to head

Model / configuration	OOF wF1	Role
ExtraTrees (800 trees)	0.87266	5% of final mixture
Logistic regression (one-hot)	0.87326	rejected — diversity candidate
MLP 128×64	0.87393	5% of final mixture
RandomForest (800 trees)	0.87432	5% of final mixture
HistGradientBoosting	0.87502	rejected
TabPFN (foundation model)	0.87644	rejected — 0.98 corr. with blend
LightGBM (single, default)	0.87646	baseline reference
CatBoost (10f $\times$ 3 seeds)	0.87871	blend member
XGBoost (10f $\times$ 3 seeds)	0.87885	blend member
LightGBM, focal loss $\gamma=1$	0.87982	best single — gain overlapped blend
GBM trio + pseudo-labels	0.87965	stage-2 core
Final: + RF/ET/MLP mixture	0.87996	submitted — private 0.880410

Negative results: rejected with evidence

Aggressive Optuna tuning

+0.002 OOF, -0.0005 leaderboard — CV overfitting via 40 trials on the same folds. Proven with a controlled probe; tuning stopped there.

Per-year thresholds

In-sample +0.001; *worse* under split-half verification — eleven fragmented optimizations, eleven chances to fit noise.

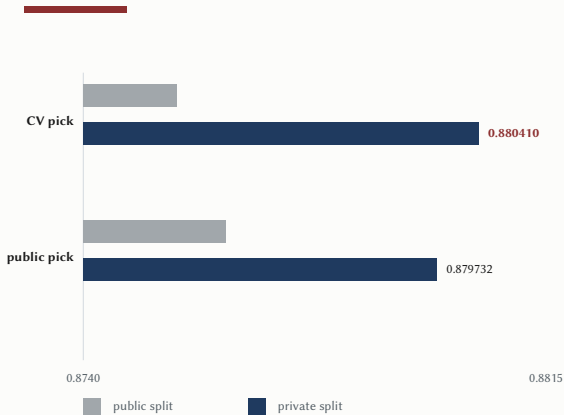
Soft-label distillation

+0.001 CV was the teacher leaking validation information through soft labels; leaderboard flat. The CV gain was provably spurious.

More model families

Everything correlated ≥ 0.97 with the blend. The dataset's signal saturates near OOF ≈ 0.880 — a measured ceiling.

The decisive call: public vs. private



- Measured: public deltas < 0.0008 between our submissions were split noise.
- Strategy: one final by public score, one by honest CV.
- The CV pick — *punished* by the public board — scored **0.880410** and delivered 4th.
- Auto-selection would have finished ≈ 9 th.

Section 6

Best Findings

6

Final results: the CV estimate was the truth

0.87996

OOF forecast (pre-reveal)

0.880410

private result

0.0004

forecast error

30th → 4th

public rank → private rank

0.000075

gap to 3rd ($\approx$ 3 patients of 36,000)

Several teams above us on the public board fell out of the private top ten — the signature of public-split overfitting.

Four principles that decided this result

1. **Identify the exact metric before modelling.** One OOF sweep revealed weighted F1 and made the threshold a model parameter.
2. **Trust out-of-fold CV; treat the public leaderboard as an instrument.** Controlled single-variable probes; never chase it.
3. **Verify every gain adversarially.** Split-half testing killed three seductive “improvements” that were bias or leakage.
4. **Log everything — negative results compound.** A complete experiment ledger made the endgame a calculation, not a gamble.

Limitations & future work

- **Censoring is implicit, not modelled** — a survival-analysis formulation (Cox PH, discrete-time hazard, boosted survival trees) is the statistically superior next step.
- **Per-era probability calibration** could sharpen the single global threshold.
- **Interpretability**: SHAP attribution of the final ensemble remains future work.
- **Compute-bounded**: four days on one laptop bounded ensemble size and search depth.

Tools: Python 3.12, pandas, scikit-learn, LightGBM, XGBoost, CatBoost, Optuna. All seeds fixed; the submitted notebook re-runs end-to-end in ≈ 1 hour and regenerates our selected predictions exactly. No external data used.

Thank you.

Questions welcome — every claim is reproducible from our notebook and experiment ledger.

Team NeurAps · Insight 2.0 — Datathon 2026 · private weighted F1
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